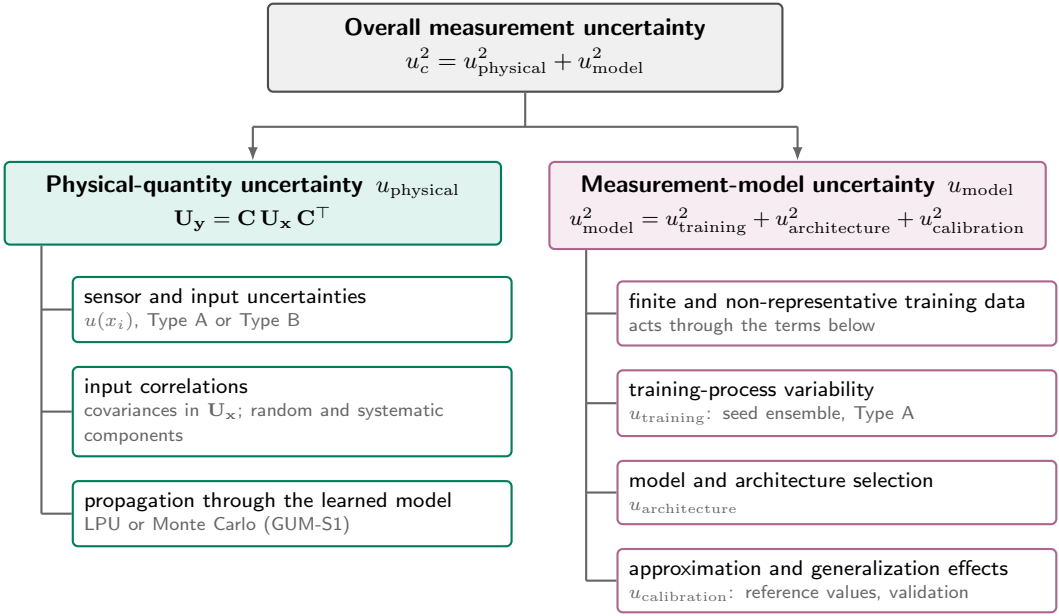

Overall measurement uncertainty
$u_c^2 = u_{\text{physical}}^2 + u_{\text{model}}^2$
Physical-quantity uncertainty u_{physical}
$\mathbf{U_y} = \mathbf{C}\,\mathbf{U_x}\,\mathbf{C}^\top$
sensor and input uncertainties
$u(x_i)$, Type A or Type B
input correlations
covariances in $\mathbf{U_x}$; random and systematic components
propagation through the learned model
LPU or Monte Carlo (GUM-S1)
Measurement-model uncertainty u_{model}
$u_{\text{model}}^2 = u_{\text{training}}^2 + u_{\text{architecture}}^2 + u_{\text{calibration}}^2$
finite and non-representative training data
acts through the terms below
training-process variability
u_{training}: seed ensemble, Type A
model and architecture selection
$u_{\text{architecture}}$
approximation and generalization effects
$u_{\text{calibration}}$: reference values, validation